

Lab 5 - Clinical Data Extraction

- Due Mar 2 by 8am
- Points 6
- Submitting an external tool
- Available Jan 12 at 8am - Mar 2 at 8am

Lab 5 - Clinical Data Extraction

LLM RAG-Based Question Answering from FHIR Data

Before attempting this lab you should watch the Lab 5 Demo video.

Overview

This lab is designed for students to experiment with using LLM RAG-based question answering from FHIR (Fast Healthcare Interoperability Resources) data. Students will be extracting clinical notes from FHIR resources, loading the notes into a vector database (Chroma DB), and using a Language Learning Model (LLM) for inference to answer user queries.

Students will...

- ...use the Google Flan-T5 model to answer specific clinical questions.
- ...compare the generated answers with pre-generated OpenAI GPT-3.5 Turbo model answers as aground truth.
- ...evaluate the quality of the answers provided by the Flan-T5 model.

No prior knowledge of machine learning and natural language processing is required. The slide presentation video explains everything needed to complete this lab.

Learning Objectives

- Understand how to use AI models to answer clinical questions from FHIR-based patient data.

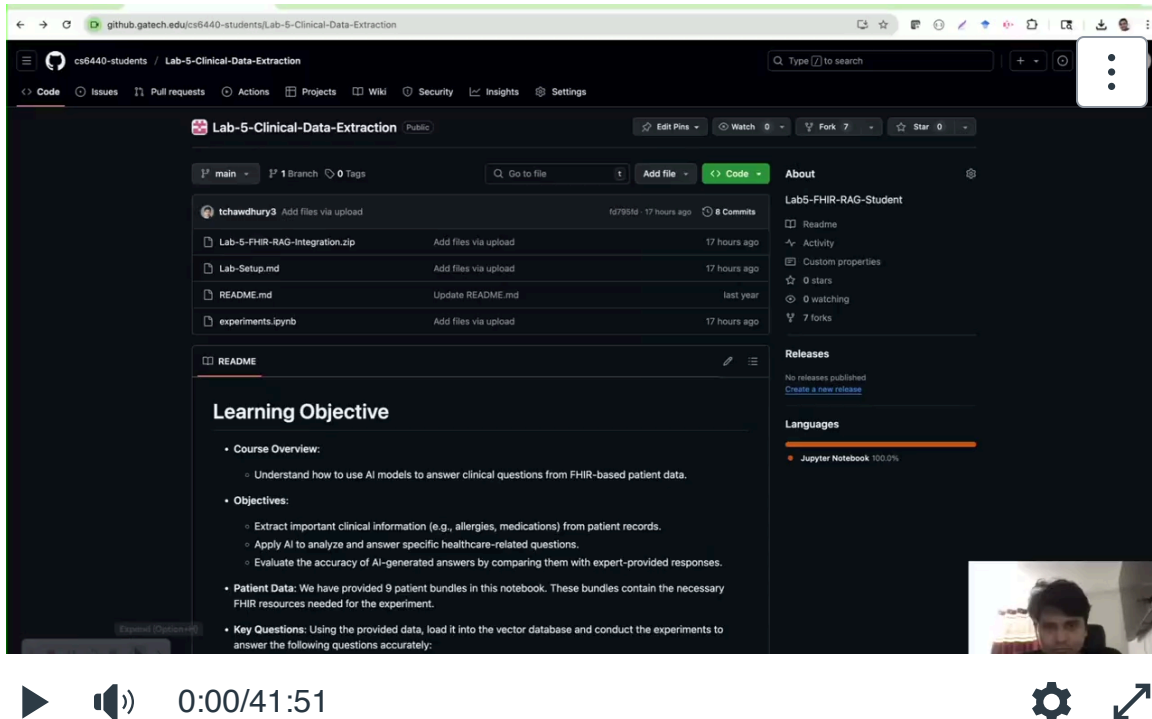
Tools Used

- [Google Colab \(https://colab.research.google.com/\)](https://colab.research.google.com/) for executing the provided notebook from the Lab 5 GitHub repository.
- [Lab 5 GitHub Repository](https://github.gatech.edu/cs6440-students/Lab-5-Clinical-Data-Extraction)  [_ \(https://github.gatech.edu/cs6440-students/Lab-5-Clinical-Data-Extraction\)](https://github.gatech.edu/cs6440-students/Lab-5-Clinical-Data-Extraction)
- Jupyter Notebook

- Google Flan-T5 Model
- OpenAI GPT-3.5 Turbo

Full Walkthrough – Don't Miss It!

Watch the complete demo to see every step of **Lab 5 Clinical Data Extraction** in action.



The screenshot shows a GitHub repository page for 'Lab-5-Clinical-Data-Extraction' by user 'tchawdhury3'. The repository is public and has 7 forks and 0 stars. The main content area displays the README file, which includes a 'Learning Objective' section. The README text is as follows:

Learning Objective

- Course Overview:
 - Understand how to use AI models to answer clinical questions from FHIR-based patient data.
- Objectives:
 - Extract important clinical information (e.g., allergies, medications) from patient records.
 - Apply AI to analyze and answer specific healthcare-related questions.
 - Evaluate the accuracy of AI-generated answers by comparing them with expert-provided responses.
- Patient Data: We have provided 9 patient bundles in this notebook. These bundles contain the necessary FHIR resources needed for the experiment.
- Key Questions: Using the provided data, load it into the vector database and conduct the experiments to answer the following questions accurately:

The video player overlay at the bottom shows a play button, a volume icon, a timestamp of 0:00/41:51, and a settings icon. A small video feed of a person is visible in the bottom right corner of the video player.

Quick Learning, Big Impact!

Tune in to this short podcast for the key objectives and simple steps you need to ace **Lab 5 Clinical Data Extraction**.



** We have also provided an option to run this Lab without Google Colab, along with instructions for local setup, which you can find here: <https://github.gatech.edu/cs6440-students/Lab-5-Clinical-Data-Extraction/blob/main/Lab-Setup.md>  (<https://github.gatech.edu/cs6440-students/Lab-5-Clinical-Data-Extraction/blob/main/Lab-Setup.md>).

Please note that this Lab was tested on an Intel MacBook with 64GB RAM. It is highly compute- and memory-intensive, and using a GPU is strongly recommended for faster performance. For this reason, we strongly recommend using Google Colab to complete the assignment.

That said, advanced students with powerful machines are more than welcome to experiment with a local setup. However, please keep in mind that we may not be able to provide the same level of support in such cases, as hardware configurations may differ.

Getting Started

Instructions

Please watch the lab video and follow the Jupyter notebook for all instructions. The lab repository with the necessary files can be found at <https://github.gatech.edu/cs6440-students/Lab-5-Clinical-Data-Extraction>  (<https://github.gatech.edu/cs6440-students/Lab-5-Clinical-Data-Extraction>).

Answering Patient Related Questions

Nine (9) patient bundles are included in the lab Jupyter notebook. These bundles contain the FHIR resources necessary for the experiment. The resources are synthetic data generated by Synthea and provided by Mitre. (You can read more about Synthea at <https://synthea.mitre.org/>  (<https://synthea.mitre.org/>) or on their GitHub repository.)

Using the provided data, you will load it into the vector database and conduct the experiments given to answer the following questions as accurately as possible:

1. What allergies does the patient have?
2. What medications is the patient currently taking?
3. What immunizations has the patient received?
4. What is the patient's primary diagnosis?
5. What procedures were conducted on the patient?
6. What treatments have been administered during the recent visit?
7. What is the patient's history of present illness?
8. What are the patient's vital info?
9. What findings are mentioned in the patient's assessment and plan?
10. What follow-up care is planned for the patient?

Supplemental Information

FHIR Bundles

- **FHIR Bundle Definition:** A FHIR Bundle is a container for a collection of resources. It is used to group multiple resources together in a single document, facilitating the management, processing, and exchange of related healthcare information. Bundles serve various purposes such as transactional operations, document composition, messaging, and more. For more information, please review the FHIR Introductory material and the FHIR Specification.
- **Structure of a FHIR Bundle:** A FHIR Bundle consists of several key elements:
 - `resourceType`: Indicates that the resource is a Bundle.
 - `id`: A unique identifier for the Bundle.
 - `type`: Specifies the purpose of the Bundle (e.g., document, transaction, searchset).
 - `entry`: An array of resources included in the Bundle.

Example of a FHIR Bundle

```

{
  "resourceType": "Bundle",
  "id": "bundle-example",
  "type": "searchset",
  "entry": [
    {
      "fullUrl": "http://example.org/fhir/Patient/123",
      "resource": {
        "resourceType": "Patient",
        "id": "123",
        "name": [
          {
            "use": "official",
            "family": "Doe",
            "given": ["John"]
          }
        ]
      }
    },
    {
      "fullUrl": "http://example.org/fhir/Observation/456",
      "resource": {
        "resourceType": "Observation",
        "id": "456",
        "status": "final",
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "3141-9",
              "display": "Body weight"
            }
          ]
        },
        "subject": {
          "reference": "Patient/123"
        },
        "valueQuantity": {
          "value": 85,
          "unit": "kg"
        }
      }
    }
  ]
}

```

This example illustrates a "searchset" Bundle containing a **Patient** resource and an **Observation** resource. Each entry includes a **fullUrl** for the resource and the resource itself. The **fullUrl** can take several forms in a valid FHIR Bundle. Please note that while we show a "searchset" type Bundle here, the lab itself uses "transaction" type Bundles.

- **Purpose of Bundling Resources**

1. **Atomic Operations:** Allows multiple operations to be grouped and executed as a single atomic unit, ensuring all operations succeed or none are applied (e.g., transaction bundles).
2. **Coherent Data Sets:** Bundles provide a way to package related resources together, making it easier to manage and transfer coherent sets of data (e.g., a clinical document).
3. **Efficient Data Exchange:** Bundling multiple resources reduces the number of individual API calls required, improving efficiency in data exchange.

4. **Complex Workflows:** Supports complex workflows by grouping all necessary resources, facilitating processing and interpretation (e.g., messaging bundles).

- **Advantages of Using Bundles in FHIR**

1. **Standardization:** Bundles provide a standardized way to group and exchange multiple resources, ensuring consistency and interoperability across different systems.
2. **Transaction Integrity:** In transaction bundles, the ability to roll back changes if any operation fails ensures data integrity and reliability.
3. **Simplified Data Retrieval:** Searchset bundles simplify the retrieval of search results, packaging them in a single response.
4. **Ease of Data Management:** Bundles make it easier to manage and transport large sets of related data, supporting operations like bulk data export and import.
5. **Enhanced Communication:** Messaging bundles facilitate structured communication between systems, ensuring that all relevant information is included and interpreted correctly.
6. **Resource Organization:** Bundles help in organizing and maintaining related resources together, providing a clearer and more manageable structure for complex data sets.

- **Types of FHIR Bundles**

- **Document:** A document bundle is used to compose a clinical document. It typically includes resources like `Patient`, `Encounter`, `Observation`, and `Medication`.
- **Message:** A message bundle is used for messaging purposes, allowing the exchange of information between systems. It contains a `MessageHeader` resource and a set of related resources.
- **Transaction:** A transaction bundle allows multiple resources to be processed as a single atomic transaction.
- **Transaction-Response:** Provides the response to a transaction bundle, including the status and results of each operation.
- **Batch:** A batch bundle allows independent processing of each operation within it.
- **Batch-Response:** Provides the response to a batch bundle, similar to a transaction-response bundle, but with independent operations.
- **History:** Represents a set of changes to resources over time, showing different versions and updates.
- **Searchset:** Contains the results of a search query, used to return a collection of resources that match the search criteria.
- **Collection:** A simple grouping of resources that are not intended to be processed together.

Summary of Key Points

- FHIR Bundles are containers for grouping multiple resources together to facilitate management, processing, and exchange of healthcare data.
- Different types of bundles serve various purposes, such as document composition, messaging, transactional operations, batch processing, and more.

- For clinical summarization and question answering, the searchset bundle is recommended due to its ability to efficiently return search query results and organize relevant resources in a single response.
- The use of FHIR Bundles enhances interoperability, efficiency, and consistency in healthcare data exchange, leading to improved patient care and data management.

Further Reading and Resources

For more detailed information on FHIR Bundles and related topics, consider the following resources:

- [HL7 FHIR Specification](https://www.hl7.org/fhir/) ➦ (https://www.hl7.org/fhir/)
- [FHIR Bundles Overview](https://www.hl7.org/fhir/bundle.html) ➦ (https://www.hl7.org/fhir/bundle.html)
- [FHIR Search API](https://www.hl7.org/fhir/search.html) ➦ (https://www.hl7.org/fhir/search.html)
- [Understanding FHIR Transactions](https://www.hl7.org/fhir/http.html) ➦ (https://www.hl7.org/fhir/http.html)
- [FHIR Clinical Documentation](https://build.fhir.org/ig/HL7/US-Core/clinical-notes.html) ➦ (https://build.fhir.org/ig/HL7/US-Core/clinical-notes.html)

These resources provide comprehensive insights into FHIR standards, implementation guides, and practical examples to further enhance your understanding and application of FHIR Bundles in healthcare interoperability.

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Submission Method

 Upload

Drag & Drop

Any file(s) including .zip. Click to browse.